

Multi-Objective Preference Inference from Demonstration

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Project Summary

In multi-objective decision making problems, eliciting preferences for a user is difficult and critical for assigning an appropriate policy. Existing approaches for inferring preferences from demonstrations either require expensive per-inference RL training (IRL-based approaches [1, 2]) or throw away important action-level information by reducing demonstrations to cumulative returns (DWPI [3]). I propose three new approaches: The first two are modified versions of DWPI, using successor features and learned trajectory embeddings respectively as richer representations of trajectories. I expect that they will both serve as a better basis for the DWPI preference inference network. My third approach is a novel algorithm, Bayesian Inverse Preference Inference (BIPI), which iteratively updates a belief distribution over preference weights by looking at the action taken at each timestep of the demonstration(s). It uses a precomputed convex coverage set (CCS) of softmax Pareto optimal policies and assumes of a Boltzmann-rational demonstrator.

Work Completed

Literature Review: I have completed a review of existing literature and discussed ideas for multi-objective preference inference from demonstrations at length with Bruno, who has done a lot of MORL research.

Use DWPI Codebase: I have cloned and successfully run the DWPI codebase. I identified what sections need to be changed to change the input to the preference inference feed-forward network.

BIPI Formalization: I made two different formalizations of BIPI. My original formulation does not have a softmax assumption but relied on having $Q_{\pi^*}(s, a) \forall a \in A(s), s \in S, \pi^* \in \Pi^*$. Where $Q_{\pi^*}(s, a)$ represents the vector of expected future return associated with following action a from state s and following the pareto optimal policy π^* . Π^* here is the set of all pareto optimal policies.

My second formulation assumes a softmax demonstrator and removes the need for those Q vectors. I made this revision since Bruno told me these Q vectors aren't typically stored when computing a CCS, and in fact may not be precisely known when using actor-critic training to find the pareto optimal policies. In this formulation I instead assume Pareto optimal policies were trained as softmax policies with known temperature β_{train} . The differences in utility ($u = w \cdot Q_{\pi^*}(s, a)$) between actions can then be expressed in terms of the optimal policies' themselves. Thus, we can express likelihood as the following:

$$P(a_t | s_t, w_n) = \frac{\pi_n^*(a_t | s_t)^\rho}{\sum_{a'} \pi_n^*(a' | s_t)^\rho}, \quad \rho = \beta_{\text{demo}} / \beta_{\text{train}} \quad (1)$$

where π_n^* is the Pareto optimal policy for candidate weight w_n . Here, candidate weights that correspond to the same optimal policy are updated by the same amount so I can discretize the preference space by partitioning the space into K regions (one per policy in the CCS) and only looking at 1 candidate policy per region. I only plan on implementing this second formulation.

Hypotheses

Hypotheses will be tested on the three benchmark environments used in the DWPI paper (Deep Sea Treasure, Traffic, Item Gathering). I will evaluate performance of 5 approaches: DWPI with return-based DWPI, state frequency-based DWPI, successor feature-based DWPI, embedding-based DWPI and BIPI. I will use the following metrics to evaluate each approach: KL-divergence and MSE of inferred vs. true weights, and utility (ground truth) under the optimal policy corresponding to the inferred preference.

1. Successor feature-based DWPI will outperform standard DWPI (based on returns and state frequencies), especially in stochastic environments.
2. Embedding-based DWPI will outperform DWPI based on returns, state frequencies and successor features, especially in stochastic environments.
3. BIPI will outperform all DWPI approaches.

I do not yet have any code or results for these hypotheses set but have formalized the approaches they are based on.

Surprises and Changes

My original proposal focused on building on top of standard DWPI, trying to detect and hedge against misspecification. After doing more reading and speaking with Bruno I became aware of DWPI's limitations. So, I pivoted and focused on developing better preference inference methods. As the work completed section shows my BIPI formalization changed a lot so that it could work with the data typically saved when calculating a CCS in MORL.

Questions and Stuck Points

What methods are typically used to create embedding of trajectories? What would make sense for this setting?

It may be a good idea to add another multi-objective environment for testing, a highly stochastic one, but I haven't found one yet. I just want to make sure the environments used in the DWPI paper aren't overly-friendly to DWPI's simplification/limitation.

I'm using Boltzmann rationality but am pretty unsure about how to handle/set the beta parameter in a principled way.

Timeline

<i>Complete</i>	Literature review, DWPI code running, BIPI formalization
April 14	Implement DWPI with successor features
April 17	Implement BIPI (new version with softmax assumption)
April 20	Find efficient way to embed trajectories
April 24	Implement embedding-based DWPI
April 29	Run experiments and compare all approaches
May 5	Final write-up

- [1] Ikenaga, A. and Arai, S. "Inverse RL approach for elicitation of preferences in multi-objective sequential optimization." *IEEE ICA*, 2018.
- [2] Takayama, N. and Arai, S. "Multi-objective deep inverse RL for weight estimation of objectives." *Artificial Life and Robotics*, 2022.
- [3] Lu, J., Mannion, P., and Mason, K. "Inferring preferences from demonstrations in multi-objective RL." *Neural Computing and Applications*, 2024.